

1. Ranking From slowest to fastest growth
 $O(1) < O(\log n) < O(n) < O(n \log n) < O(n^5)$

2. Big-O of the given functions.

(a) $4n^2 + 20n + 9$

Dominant term is n^2

$O(n^2)$

(b) $25n^3 + 15$

Dominant term is n^3

$O(n^3)$

(c) $n^3 + 5 \log n + 6$

Dominant term is n^3

$O(n^3)$

(d) $10^5 + 10^5 n + 10^5 n^2$

10^5 is constant Dominant term is n^2 .

$O(n^2)$

(e) $72n(n + 3n^2)$

$72n^2 + 216n^3$

$O(n^3)$

(f) $2^8 + 32^4 n - 5$

$2^8 = 256$ and $32^4 = 1048576$ are constants

$1048576n + 251$

$O(n)$

3. $O(n)$ algorithm to Sum 1 to n

Sum := 0

for $i := 1$ to n do:

Sum := Sum + i

endfor

return Sum

The loop executes n times, each iteration $O(1)$, so
Worst-case $O(n)$.

4. Worst-case running time

a.

$X = 4$

for $i := 0$ to number do:

$X := X + 1$

endfor

Loop runs number + 1 times.

$O(\text{number})$ is $O(n)$.

b. choice := get-choice() // $O(1)$.

If choice = 1 then;

Print "Ready"

else if choice = 2 then;

for $i := 0$ to Cycles do:

Print i

endfor

endif.

Best case choice = 1 is $O(1)$. Worst case choice = 2
runs the loop.

Worst-case $O(\text{cycles})$.

5. Fill in the blanks

① For some function $f(n)$ and $g(n)$, if $f(n)$ is $O(g(n))$, then for some constants C and K , for all $n > K$, $f(n) \leq C g(n)$.

② For some function $f(n)$ and $g(n)$, if $f(n)$ is $\Omega(g(n))$, then for some constants C and K for all $n > K$, $f(n) > C g(n)$.

6. Show $4n+12$ is $O(n)$

We need $C, K > 0$ such that $4n+12 \leq Cn$ for all $n > K$.

choose $C=5$

$$4n+12 \leq 5n$$

$$12 \leq n$$

So Take $K=12$

Then for all $n > 12$, $4n+12 \leq 5n$

Hence $4n+12 = O(n)$ with $C=5$, $K=12$

Any $C > 4$ work with $K > 12/(C-4)$.

7. Show $5n^2-16$ is $O(n^2)$

We need $5n^2-16 \leq cn^2$ for $n > K$

Take $C=5$:

$5n^2-16 \leq 5n^2$ holds for all $n > 0$.

So, $C=5$, $K=4$ (or $K=0$) is Valid.

If we insist on $C=1$, the inequality $5n^2-16 < n^2$ gives $4n^2 \leq 16$ i.e. $n \leq 2$, which does not give an eventual upper bound, Hence C must be > 5

A convenient choice is $C=5$, $K=1$.

$$\therefore 5n^2-16 = O(n^2).$$